

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250270804

Kind Code

A1

Publication Date

August 28, 2025

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METHOD AND ARRANGEMENT FOR COUPLING A FOUNDATION STRUCTURE AND A VERTICAL PRECAST STRUCTURE

Abstract

A connection assembly mechanism includes a threaded connector, a hollow box section member, a set of elongated connectors, and a connection mechanism. The threaded connector is embedded in a foundation structure. The hollow box section member having a base including a threaded cavity receives the threaded connector and is flush mounted with an internal bottom edge of a vertical precast structure. The set of elongated connectors, positioned at opposite vertical sides of the hollow box section member, is embedded within the vertical precast structure. The connection mechanism coupling the hollow box section member and the threaded connector, thereby couples the vertical precast structure and the foundation structure.

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Family ID: 1000007708858

Appl. No.: 18/584823

Filed: February 22, 2024

Publication Classification

Int. Cl.: E04B1/21 (20060101); E04B1/41 (20060101)

U.S. Cl.:

CPC E04B1/21 (20130101); E04B1/4157 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure generally relates to construction technology. In particular, the present disclosure relates to a method and an arrangement for coupling a foundation structure and a vertical precast structure for use in construction technology.

BACKGROUND OF THE INVENTION

[0002] Existing construction technologies involve one-off (e.g., customized) build-on site approaches in which construction material is brought to the construction site where the actual construction is performed. This has been the traditional methodology and approach for many years but has certain inherent challenges, including non-availability of skilled workforce (e.g., manual labor), heavy and expensive on-site machinery, incorrect estimate of completion time of construction projects, delays in delivery of projects, inclement weather, poor quality and wastage of materials, noise and air pollution, and cost involved in disposal of debris. This approach is also "one-off" as it provides no repeatability or scalability leverage. Each building is constructed, and each project is performed differently, and results vary widely, which may be undesirable considering present day demand for symmetrical construction projects with enhanced look and feel. However, constructing or casting each individual component of a building on site incurs significant expenditures in time and resources. It also increases a project's vulnerability to unforeseen factors, such as poor weather, worksite accidents, improper pour, etc.

[0003] Such traditional methodology and approach use connection methods that typically include placing rebars and lapping them together with adjacent bars during formwork. However, such connection methods are time and manpower intensive during installation. Further, on-site rebar binding makes such connection methods quite cumbersome to handle. Besides, substantial waiting time has to be allowed for using such connection methods to gain sufficient strength before starting the next level of work.

[0004] In order to address the aforesaid shortfalls of such build or cast-on site approaches, some construction projects use precast modules. Examples of the precast modules may include walls, beams, slabs, and the like, that are built in factories under factory scaling, repeatability, and in-factory conditions. The precast modules are then delivered to a building site and installed using standard connection mechanisms. Such standard connection mechanisms take less time as compared to the connection methods for the build-on site approaches.

[0005] However, even standard connection mechanisms have different challenges for different types of coupling mechanisms. In an example for foundation-to-wall connections, skilled labor is required for non-shrink grouting and pressure grouting. Further, waiting time needs to be allowed until the grout, with a color different from concrete color, achieves sufficient strength.

[0006] Accordingly, there remains a need in the art for an improved connection assembly mechanism that not only requires lesser skilled workers on site and saves cost and time in the installation process, but also ensures the structural continuity of the whole structure and transfer of forces between the prefabricated building modules during ultimate load conditions.

SUMMARY OF THE INVENTION

[0007] Embodiments for a connection assembly mechanism for coupling a foundation structure and a vertical precast structure in construction technology are disclosed that address at least some of the above challenges and issues.

[0008] In an aspect, the present disclosure is directed to a connection assembly mechanism. The connection assembly mechanism includes a threaded connector configured for embedding in a foundation structure. The connection assembly mechanism further includes a hollow box section member having a base including a threaded cavity configured for receiving the threaded connector,

the hollow box section member configured for flush mounting with an internal bottom edge of a vertical precast structure. The connection assembly mechanism further includes a set of elongated connectors positioned at opposite vertical sides of the hollow box section member, wherein each of the elongated connectors is configured for embedding within the vertical precast structure. The connection assembly mechanism further includes a connection mechanism coupling the hollow box section member and the threaded connector, for thereby coupling the vertical precast structure and the foundation structure.

[0009] In an embodiment of the present disclosure, the hollow box section member and the threaded connector are mechanically coupled using a mechanical fastener for coupling the vertical precast structure with the foundation structure at a construction site.

[0010] In an embodiment of the present disclosure, the connection assembly mechanism further includes a plate member with a cavity adaptable to receive the threaded connector. The plate member is configured to distribute a load under the mechanical fastener when the vertical precast structure is affixed with the foundation structure at the construction site.

[0011] In an embodiment of the present disclosure, the hollow box section member is flush-mounted with the internal bottom edge of the vertical precast structure such that the hollow box section member is accessible for the mechanical coupling with the foundation structure from an internal side of the vertical precast structure.

[0012] In an embodiment of the present disclosure, the threaded connector corresponds to a cast-in-place anchor bolt.

[0013] In an embodiment of the present disclosure, the threaded connector corresponds to a post-installed anchor bolt set onto anchoring adhesive.

[0014] In an embodiment of the present disclosure, the foundation structure corresponds to a monolithic cast-in-place foundation structure.

[0015] In an embodiment of the present disclosure, the connection assembly mechanism further includes a sealant sealing a gap between the hollow box section member flush-mounted with the internal bottom edge of the vertical precast structure and a top surface of the foundation structure.

[0016] In an aspect, the present disclosure is directed to a connection method that includes anchoring a threaded connector in a foundation structure, and flush-mounting a hollow box section member with a bottom surface of a vertical precast structure. A set of elongated connectors, positioned at opposite vertical sides of the hollow box section member, is embedded within the vertical precast structure. A base of the hollow box section member includes a threaded cavity adaptable to receive the threaded connector. The connection method further includes coupling the hollow box section member and the threaded connector using a connection mechanism for coupling the vertical precast structure with the foundation structure.

[0017] In an embodiment of the present disclosure, the connection method further includes determining a type, quantity, and size of a connection assembly mechanism for coupling the foundation structure and the vertical precast structure based on a first set of parameters associated with the connection assembly mechanism and a second set of parameters associated with at least one of the foundation structure and the vertical precast structure.

[0018] In an embodiment of the present disclosure, the first set of parameters includes material specifications of the connection assembly mechanism.

[0019] In an embodiment of the present disclosure, the material specifications of the connection assembly mechanism correspond to at least tensile strength and hardness of the connection assembly mechanism. The second set of parameters includes at least one of a location, an orientation, and a weight of at least one of the foundation structure and the vertical precast structure and environmental conditions.

[0020] In an embodiment of the present disclosure, the hollow box section member and the threaded connector are mechanically coupled using a mechanical fastener for coupling the vertical precast structure with the foundation structure at a construction site.

[0021] In an embodiment of the present disclosure, the hollow box section member further comprises a plate member with a cavity that is adaptable to receive the threaded connector. The plate member is configured to distribute a load under the mechanical fastener when the vertical precast structure is affixed with the foundation structure at the construction site.

[0022] In an embodiment of the present disclosure, the hollow box section member is flush-mounted with the internal bottom edge of the vertical precast structure such that the hollow box section member is accessible for the mechanical coupling with the foundation structure from an internal side of the vertical precast structure.

[0023] In an embodiment of the present disclosure, the threaded connector corresponds to a cast-in-place anchor bolt.

[0024] In an embodiment of the present disclosure, the threaded connector corresponds to post-installed anchor bolt set onto anchoring adhesive.

[0025] In an embodiment of the present disclosure, the foundation structure corresponds to a monolithic cast-in-place foundation structure.

[0026] In an embodiment of the present disclosure, the method further includes sealing a gap between the hollow box section member flush-mounted with the internal bottom edge of the vertical precast structure and a top surface of the foundation structure with a sealant.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] Further advantages of the disclosure will become apparent by reference to the detailed description of preferred embodiments when considered in conjunction with the drawings. In the drawings, identical numbers refer to the same or a similar element.

[0028] FIG. 1 is a perspective view of an enclosure structure, in accordance with some embodiments.

[0029] FIG. 2A is an enlarged, detailed view of a connection assembly mechanism, in accordance with some embodiments.

[0030] FIG. 2B is a side view of the connection assembly mechanism coupling the foundation structure and the first wall structure, in accordance with some embodiments.

[0031] FIG. 3 illustrates a block diagram of an exemplary system in accordance with some embodiments.

[0032] FIG. 4 illustrates the steps of a method of coupling a wall structure and a foundation structure in accordance with some embodiments.

DETAILED DESCRIPTION

[0033] The following detailed description is presented to enable any person skilled in the art to make and use the disclosure. For purposes of explanation, specific details are set forth to provide a thorough understanding of the present disclosure. However, it will be apparent to one skilled in the art that these specific details are not required to practice the disclosure. Descriptions of specific applications are provided only as representative examples. Various modifications to the preferred embodiments will be readily apparent to one skilled in the art, and the general principles defined herein may be applied to other embodiments and applications without departing from the scope of the disclosure. The present disclosure is not intended to be limited to the embodiments shown but is to be accorded the widest possible scope consistent with the principles and features disclosed herein.

[0034] With modernization in construction-related methodologies and technologies, there has been a rapid shift from normal customized build on-site construction methodologies to construction using modules or blocks that may be built off-site and then assembled on-site to form a construction or a building. However, in the latter case, suitable connection assemblies for each use,

under different environmental and ultimate load conditions are of utmost importance. The role of a connection assembly mechanism is not only to fix the prefabricated structures together, but also to ensure the structural continuity of the whole structure and to transfer forces between the prefabricated structures when the system is loaded.

[0035] Conventional connection methods for build or cast-on site approaches may be too time consuming and labor intensive for execution. Further, on-site rebar binding makes such connection methods quite cumbersome to handle. Furthermore, substantial waiting time has to be allowed while using such connection methods for foundation and wall elements to gain sufficient strength before starting the next level of work.

[0036] Further, standard connection mechanisms for standard precast modules are also not able to address the aforesaid shortfalls of conventional connection methods. By way of an example, the standard connection mechanisms require skilled labor for non-shrink grouting and pressure grouting for foundation-to-wall connections, and the like. Further, standard connection mechanisms demand waiting time till connection or grout strength is achieved.

[0037] Clearly, current connection assemblies and technologies fail to address many concerns, as described hereinbefore. This is especially true for advanced precast structures, which are lightweight due to less material requirements, are more economical (due to less labor, accelerated manufacturing, and reduced cost), and have better performance, factors that together meet today's requirements of enhanced, efficient, and robust connection assemblies.

[0038] The embodiments of the present disclosure address these concerns by providing improved and better-quality connection assemblies, requiring lesser-skilled workers on site and saving cost and time in the installation process. Further, such connection assemblies also ensure the structural continuity of the whole structure and the transfer of forces between the prefabricated structures during ultimate load conditions.

[0039] The disclosed architecture/solution provides several other objects and advantages, some of which are discussed below. The present disclosure supports rapid construction of a structure including precast (prefabricated) modules accommodating erratic site constraints/conditions and/or tight construction schedules. Further, the present disclosure provides at least pre-compression forces across such connection assemblies, thus improving the durability of a building structure. Additionally, an increase in the load resistance of the structure may be obtained by means of embodiments in accordance with the present disclosure. At the least, this durability makes the structure economical and also helps reduce the construction times.

[0040] Connection assemblies in accordance with the embodiments provide advantages in their simplicity of manufacture and system performance, and hence require little or no future maintenance. By leveraging a controlled environment production, these connection assemblies ensure quality, cost reduction, and speedy installation.

[0041] Further, the connection assembly mechanism in accordance with the present disclosure exhibits substantial fire resistance owing to metallic body, which in turn reduces insurance costs due to increased safety, security, reliability, and structural soundness.

[0042] Structurally, the connection assembly mechanism in accordance with the present disclosure provides substantial loading capacity and further offers the building structure a very high resistance to wind, hurricanes, floods, and other damaging environmental occurrences.

[0043] Certain terms and phrases have been used throughout the disclosure and will have the following meanings in the context of the ongoing disclosure.

[0044] "Cast-in-place concrete" or "Cast-in-situ concrete" is a building-construction technology where elements of a building structure are cast at the site in formwork.

[0045] "Pre-cast structure" refers to a construction product produced by casting concrete in a reusable mold or form which is then cured in a controlled environment at an offsite location, transported to a construction site, and maneuvered into a targeted place. Examples may include pre-cast beams, slabs, wall panels, and the like.

[0046] “Wall panel” refers to a prefabricated multi-layered wall fabricated at an offsite location and installed on-site, wherein “on-site” denotes a construction site and “offsite” denotes away from the construction site. The wall panel may be with or without door or window openings based on the design of the building structure.

[0047] “Slab” refers to a prefabricated structure formed at an offsite location and installed on-site. A slab includes a concrete base and a structural topping. The structural toppings, also referred to as topping screeds, are specialized materials applied to an existing concrete base of the slab to enhance performance, durability, and aesthetics.

[0048] “Beam” refers to a construction element that is made by casting concrete into a mold and then curing it in a controlled environment at the offsite location. Beams may be used for both load-bearing and non-load bearing applications and may be made in a variety of shapes and sizes.

[0049] “Foundation” refers to a monolithic cast-in-place foundation that is poured in one piece, typically 4-6 inches thick. It is commonly used in areas with shallow soil frost depths and is known for its durability and efficiency.

[0050] “Grout” is a filling, which when poured into a receptacle will fill in the receptacle and consolidate the adjacent edges into a solid mass, such as cementitious mortar or other cement-based materials, bentonite, bentonite/sand mixtures, graphite-based materials, carbon nanotubes and nanofibers, or similar materials.

[0051] “Sealant” refers to substances used to seal, block, or close gaps between various modules of the building structure to prevent fluids, air, and pests from passing through. These materials seal joints where dissimilar materials meet, filling gaps regardless of any irregularities that may exist between the two joint surfaces.

[0052] “Backer Rod” refers to a backing in joints. It controls the sealant thickness and amount of sealant needed to fill a gap between joints. The backer rod forces the sealant to the sidewalls to ensure contact and proper adhesion of the sealant.

[0053] “Modular” refers to individual and independent blocks or any mechanism or procedure for arranging them.

[0054] “Lattice girders” are three-dimensional, industrially prefabricated reinforcing elements. They consist of an upper chord, two lower chords, and continuous truss wires (e.g., diagonal chords). Typically, the continuous truss wires are connected to the chords by means of electric resistance welding.

[0055] FIG. 1 is a perspective view of an enclosure structure **100**, in accordance with some embodiments. The enclosure structure **100** includes precast structures including a foundation structure **102**, a first wall structure **104**, a second wall structure **106**, a slab structure **108**, a beam structure **110**, and a staircase slab **112**. The enclosure structure **100** further includes a connection assembly mechanism **114**. The precast structures may be prefabricated at an offsite location away from a construction site and installed on-site at the construction site.

[0056] The foundation structure **102** may correspond to a monolithic cast-in-place foundation structure. The foundation structure **102** is the lowest part of the enclosure structure **100** that is in direct contact with the soil and transfers loads from the enclosure structure **100** to the soil safely. To construct the foundation structure **102**, trenches are dug deeper into the soil till a hard stratum is reached. Reinforcement cages are incorporated, and concrete is poured. Because the foundation structure **102** is a cast-in-place module and is poured all at once, it is erected much faster, thereby keeping labor costs low. The foundation structure **102** is designed to account for the characteristics of the underlying soil and local environment (e.g., slope of underlying soil, soil type, compactness, local weather conditions, etc.) such that the underlying soil below the foundation structure **102** does not undergo shear failure.

[0057] The first wall structure **104** may correspond to a level-1 wall which is installed directly on the foundation structure **102**, whereas the second wall structure **106** may correspond to a level-2 wall which is installed directly on the first wall structure **104**. The first wall structure **104** and the

second wall structure **106** may be multi-layered precast structures that can withstand load, climate changes, and daily wear and tear as may be subjected to the enclosure structure **100**. It will be appreciated that the first wall structure **104** and the second wall structure **106** obtained due to their construction technology are high quality, forming repeatable and scalable building structures. It will be appreciated that the enclosure structure **100** is merely illustrative of some embodiments. For example, in some embodiments, the first wall structure **104** may be installed indirectly on the foundation structure **102**, such as by intermediate elements, the second wall structure may be installed indirectly on the first wall structure **104**, or both.

[0058] The slab structure **108** may correspond to a precast structure that includes a concrete base and a structural topping. The structural topping may be specialized materials applied to an existing concrete base of the slab structure **108** to enhance performance, durability, and aesthetics. In some embodiments, the slab structure **108** may be a lattice girder slab that may act as permanent formwork and as precast soffits for robust, high-capacity composite slabs. The slab structure **108** may be cast with most, if not all, of the bottom reinforcement; the top reinforcement may be fixed in situ.

[0059] The beam structure **110** may correspond to a precast horizontal structure that may be able to withstand vertical loads, shear forces, and bending moments. The beam structure **110** may transfer loads that are imposed along its length to its endpoints, such as the foundation structure **102**, the first wall structure **104**, and the like. The beam structure **110** may be used for both load-bearing and non-load bearing applications and may be made in a variety of shapes and sizes.

[0060] The staircase slab **112** may be a precast structure designed to provide a vertical access from one level to another in the enclosure structure **100**. The staircase slab **112** may be made up of reinforced concrete that eliminates the trouble of adjusting the number of steps, rise, run and width of each stair flight. The staircase slab **112** may include a single precast unit containing all the flights and landings or separate precast flights and landings.

[0061] The connection assembly mechanism **114** may correspond to various brackets and shear connectors that are embedded in different precast structures (such as the first wall structure **104**), and cast-in-place structures (such as the foundation structure **102**), which when coupled at the construction site, results in a robust installation of a base portion of level-1 of the enclosure structure **100**. The structure and installation of the connection assembly mechanism **114** is described in detail below.

[0062] Once the aforesaid precast structures, such as the first wall structure **104**, which are prefabricated, reinforced, and layered concrete elements, have been constructed in a factory setting, they are subsequently transported to the construction site where the foundation structure **102** is already in place and hardened. At the construction site, the precast structures, e.g., the first wall structure **104**, are mechanically positioned in accordance with a pre-designed layout for the enclosure structure **100** and coupled with the foundation structure **102** using multiple instances of the connection assembly mechanism **114**.

[0063] With reference to the building blocks disclosed in FIG. **1**, in some embodiments, various connection methodologies and/or technologies, such as the connection assembly mechanism **114**, are utilized to couple the foundation structure **102** and the first wall structure **104** and make the enclosure structure **100** structurally and environmentally seamless. The connection assembly mechanism **114**, as described herein, may speed up installation processes and reduce the need for skilled labor.

[0064] Since the enclosure structure **100**, as proposed by the present disclosure, is a modular construction, the precast structures are installed in an ordered manner so as to not create a decompression load, the precast structures sequentially erected vertically, increasing the height of the enclosure structure **100** as per the pre-designed layout for the enclosure structure **100**. Once the installation corresponding to the structural work has finished, the placement of doors and windows, bathroom fittings are then carried out in order for a final painting job. It may be contemplated by the

present disclosure that the connection assembly mechanism **114** may be employed in any commercial or residential structure having a variety of dimensions and stories.

[0065] FIG. 2A is an enlarged, detailed view **200A** of the connection assembly mechanism **114**, in accordance with some embodiments. FIG. 2B is a side view **200B** of the connection assembly mechanism **114** coupling the foundation structure **102** and the first wall structure **104**, in accordance with some embodiments.

[0066] Referring to FIG. 2A, there are shown a hollow box section member **202**, a set of elongated connectors **204A** and **204B**, a threaded connector **206**, a plate member **208**, and a mechanical fastener **210**. Referring to FIG. 2B, there are shown compacted fill **212**, cast-in-place (CIP) foundation footing **214A**, CIP slab on grade **214B**, slab reinforcement **216**, drilled hole **218**, anchoring adhesive **220**, recess **222**, backer rod **224**, sealant **226**, rebars **228**, and vapor barrier **230**. FIGS. 2A and 2B have been described in conjunction with each other.

[0067] Referring to FIG. 2A, there is shown the hollow box section member **202** flush-mounted with an internal bottom edge of a vertical precast structure, e.g., the first wall structure **104**. The hollow box section member **202** may be flush-mounted with the internal bottom edge of the first wall structure **104** such that the hollow box section member **202** is accessible for the mechanical coupling with the foundation structure **102** from an internal side of the first wall structure **104**. The hollow box section member **202** may be flush-mounted with the internal bottom edge of the first wall structure **104** during casting of the first wall structure **104** in a controlled environment of the factory setting. In some embodiments, a base of the hollow box section member **202** may include a threaded cavity adaptable to receive the threaded connector **206**. The locations and the spacing between various instances of the hollow box section member **202** along the bottom surface of a vertical precast structure, e.g., the first wall structure **104**, may be as per the pre-designed layout of the enclosure structure **100**, or as recommended by an automated system, such as system **300**, as described in FIG. 3.

[0068] There are also shown the set of elongated connectors **204A** and **204B** that are positioned at opposite vertical sides of the hollow box section member **202**. The set of elongated connectors **204A** and **204B** are embedded within the first wall structure **104** during casting of the first wall structure **104** in the controlled environment of the factory setting. In an exemplary embodiment, the spacing between the set of elongated connectors **204A** and **204B** may be at the most 8 inches. It will be appreciated that though only two connectors in the set of elongated connectors **204A** and **204B** have been shown, the disclosure is not so limited and more than two elongated connectors may also be possible, without any deviation from the scope of the disclosure.

[0069] The threaded connector **206** may correspond to a connector that fixedly engages with the foundation structure **102** of the enclosure structure **100**. In some embodiments, the threaded connector **206** may correspond to a cast-in-place anchor bolt member. In other words, the threaded connector **206** may be embedded in the foundation structure **102** while being cast at the construction site. In other embodiments, the threaded connector **206** may correspond to a post-installed anchor bolt set onto the anchoring adhesive **220**. In such embodiments, a correct hole size and depth may be drilled in the foundation structure **102**. The drilled hole is cleaned and the anchoring adhesive **220** may be injected from the bottom of the hole upwards. The threaded connector **206** (which may be a stud, bolt, anchor, rebar, or a threaded rod) may be inserted and rotated into the hole and left until the anchoring adhesive **220** has hardened and cured.

[0070] The locations and the spacing between adjacent instances of the threaded connector **206** along the length of the foundation structure **102** may be as per the pre-designed layout of the enclosure structure **100**. It will be appreciated that the locations and the spacing between adjacent instances of the threaded connector **206** along the length of the foundation structure **102** must conform to the locations and the spacing between adjacent instances of the hollow box section member **202** along the bottom surface of the first wall structure **104**, as the threaded cavity of each hollow box section member **202** is designed to receive corresponding threaded connectors **206** for a

proper interconnection.

[0071] In some embodiments, the hollow box section member **202** and the threaded connector **206** may be mechanically coupled using a connection mechanism for coupling the vertical precast structure, e.g., the first wall structure **104**, with the foundation structure **102**. For example, the hollow box section member **202** and the threaded connector **206** are mechanically coupled using a mechanical fastener, such as a screw or a nut, for coupling the first wall structure **104** with the foundation structure **102** at the construction site. In some embodiments, the plate member **208** with a cavity adaptable to receive the threaded connector **206** may also be used. The plate member **208** may be sandwiched between the top surface of the base of the hollow box section member **202** and base of the mechanical fastener **210**. Accordingly, the plate member **208** may distribute a load under the mechanical fastener **210** when the first wall structure **104** is affixed with the foundation structure **102** at the construction site. Though the mechanical fastener **210** is shown and described herein to be a nut member, it should be noted that the disclosure should not be limited to the nut member only, and other forms of the mechanical fastener **210** may also be used, without any deviation from the scope of the disclosure.

[0072] In some embodiments, there may be gaps between the hollow box section member **202** flush-mounted with the internal bottom edge of the first wall structure **104** and a top surface of the foundation structure **102**. Such gaps may be sealed with the sealant **226**.

[0073] Referring to FIG. 2B for further details, for site preparation of the enclosure structure **100**, an area is cleared of vegetation, debris, and any potential obstructions, and then leveled and compacted to ensure a stable and even base, referred to as the compacted fill **212**. A wooden or metal formwork may be then set up to define the shape and dimensions of the CIP foundation footing **214A** and the CIP slab on grade **214B**. The formwork may also act as a barrier, preventing the poured concrete from spreading beyond the desired area. Before pouring concrete, plumbing and electrical lines are laid out and attached to the formwork. Further, to reinforce the concrete and enhance its strength and durability, slab reinforcement **216**, rebars **228**, steel rebar or wire mesh may be placed within the formwork. In some embodiments, such reinforcement is important for handling the load and stress imposed on the foundation structure **102**. Vapor barrier **230** may also be arranged to prevent moisture from entering the CIP foundation footing **214A** and the CIP slab on grade **214B**, and to further avoid flooring problems.

[0074] Once the preparation is complete, the concrete is poured into the formwork, ensuring it is evenly distributed to achieve a uniform thickness. While pouring, a recess **222** or a pocket may be provided where the first wall structure **104** is to be installed. In some embodiments, the size of the recess **222** may be in accordance with the thickness of the first wall structure **104** and any additional indentation for assembling the connection assembly mechanism **114**. Further, after assembling, the recess **222** may further enable a worker to fill the gap underneath the bottom of the first wall structure **104** and the top surface of the foundation structure **102** by using the backer rod **224** and the sealant **226**. In some embodiments, the thickness of the first wall structure **104** may be 8 inches, and the recess may be 10 inches wide. The poured concrete may be then leveled and smoothed to create a seamless surface. It will be appreciated that in other embodiments, the first wall structure **104** and the recess may have other thicknesses and dimensions.

[0075] Once the CIP foundation footing **214A** and the CIP slab on grade **214B** are set, the threaded connector **206**, such as a bolt, may be post-anchored in the foundation structure **102** as per the pre-designed layout of the enclosure structure **100** and later allowed to cure. Thereafter, the first wall structure **104** with the hollow box section member **202** flush-mounted with the internal bottom edge may be positioned to conform (e.g., align) with the threaded connector **206** along the length of the foundation structure **102**. The threaded connector **206** may be received at the threaded cavity provided at the base of the hollow box section member **202** followed by the cavity of the plate member **208**. Once the arrangement is in place, the mechanical fastener **210**, such as a nut member, may be screwed onto the threaded connector **206** and the first wall structure **104** may be affixed

with the foundation structure **102** at the construction site. Post affixing, gaps formed underneath the bottom surface of the first wall structure **104** and the top surface of the foundation structure **102** may be fitted with the backer rod **224** that controls the thickness, spread, and amount of the sealant **226** needed to fill the gap. The backer rod **224** may be fitted both towards the internal and the external sides of the first wall structure **104**. Afterwards, the sealant **226** may be used to completely fill such gaps to ensure contact and proper adhesion.

[0076] FIG. **3** illustrates a block diagram of an exemplary system **300** in accordance with embodiments of the present disclosure. The system **300** includes a processor **302**, a memory **304**, input/output devices **306**, a network interface **308**, and a Computer-Aided Utilities (CAU) module **310** that includes an intelligent recommendation module **312** and a computer-aided manufacturing module **314**. The system **300** may be further communicatively coupled with a computer numerical control (CNC) machine **318** via a communication network **316**.

[0077] The processor **302** may comprise suitable logic, circuitry, and interfaces that may be configured to execute instructions stored in the memory **304** or commands provided by a user. The processor **302** may also collect information for processing, store it in the memory **304**, and may transmit to other modules, such as the CAU module **310** and the I/O devices **306**. In accordance with some embodiments, the computing functionalities of the processor **302** disclosed herein may be implemented in one or more silicon cores in a RISC processor, an ASIC processor, a CISC processor, FPGAs, and other semiconductor chips, processors, or control circuits.

[0078] The memory **304** may comprise suitable logic, circuitry, and interfaces that may be configured to store data supporting various functionalities performed by the processor **302** and the CAU module **310**. The memory **304** may store information and/or instructions, various application programs or applications, and a set of data and commands for various operations performed by the processor **302** and the CAU module **310**. The memory **304** may include volatile and non-volatile memory, such as a random-access memory (RAM) and a read only memory (ROM). Several program modules may be stored on the hard disk, external disk, the RAM, or the ROM, including an operating system, one or more application programs, and program data. The RAM may be of any type, such as SRAM, DRAM, or SDRAM. A BIOS containing the basic routines that may transfer information between elements within the system, such as during start-up, may be stored in the ROM.

[0079] The I/O devices **306** may include input devices, such as keyboard, mouse, microphone, camera, light pen, gesture recognition devices, scanner, touch screen and the like, that may be used to provide input to the system **300**. The input may include a layout of the enclosure structure **100**, a weight of the first wall structure **104**, a quality of the concrete, and the like. The I/O devices **306** may further include output devices, such as touch screen, printer, display screen, and the like. The output may include recommendations for the type, quantity, size, location, and spacing pertaining to multiple instances of the connection assembly mechanism **114** to be used for coupling the foundation structure **102** with the first wall structure **104**.

[0080] The network interface **308** may be configured to transmit/receive the data from the I/O devices **306** over the communication network **316** to/from other network interfaces of other devices. In some embodiments, the network interface **308** may transmit the code files to other platforms, such as the CNC machine **318**, for fabricating or manufacturing the connection assembly mechanism **114** with desired geometry, structure, and design. The network interface **308** may include wired communication interfaces, wireless communication interfaces, cellular communication interfaces, and other communication interfaces to provide communication via other modalities, known in the art.

[0081] The CAU module **310** may comprise suitable logic, circuitry, and interfaces that may be configured to perform various functionalities to intelligently handle the fabrication and/or manufacturing of the connection assembly mechanism **114** based on the requirement and the layout of a structure, such as the enclosure structure **100**.

[0082] In some embodiments, the CAU module **310** may include the intelligent recommendation module **312** programmed, for example, with a rules-based algorithm retrieved from the memory **304** and user-defined settings received from the I/O devices **306**. Accordingly, the CAU module **310** may automatically incorporate local rules, knowledge, geographical information, and content for indicating type, quantity, and sizes of different components of connection assembly mechanism **114** for coupling the foundation structure **102** with the first wall structure **104** based on a first and a second set of parameters. The types of materials for the connection assembly mechanism **114** may be based on carbon content, content based on different alloying elements, or environmentally safe content. The quantity of the connection assembly mechanism **114** may correspond to how many connection assemblies are required for coupling the foundation structure **102** with the first wall structure **104**. The size of the connection assembly mechanism **114** may refer to the dimensions of each component in proportion to each other. In some embodiments, the intelligent recommendation module **312** of the CAU module **310** may further recommend locations and the spacing between various instances of the connection assembly mechanism **114** along the first wall structure **104** and the foundation structure **102** based on the first and the second set of parameters.

[0083] The first set of parameters may include, for example, material specifications of the metal used for the connection assembly mechanism **114**. Material specifications may include, for example, tensile strength, yield strength, minimum elongation, and hardness of the connection assembly mechanism **114**. For example, if the tensile strength of the material is low, then a greater number of connection assemblies may be required and vice versa. The second set of parameters may include, for example, at least one of a location, an orientation, and a weight of the vertical precast structure, i.e., the first wall structure **104**, and various environmental conditions, such as those corresponding to hurricane sensitive or earthquake sensitive zones. For example, if the enclosure structure **100** is in a hurricane sensitive zone, then a greater number of connection assemblies of higher tensile strength may be required. Thus, the intelligent recommendation module **312** may automatically apply local rules to conform with regulations, building codes, local preferences, and user-defined settings for the manufacture, fabrication, and/or installation of the connection assembly mechanism **114**.

[0084] The intelligent recommendation module **312** may save the information relating to the recommendation of the connection assembly mechanism **114** to computer files that may be stored in the memory **304**.

[0085] In some embodiments, the computer-aided manufacturing module **314** may translate the information, relating to the recommendation of the connection assembly mechanism **114** corresponding to the fabrication and installation of the connection assembly mechanism **114**, to a machine tool code language. By way of non-limiting example, G-Code, based on the RS-274 standard, may be used as a machine tool code language. The computer-aided manufacturing module **314** may save the information relating to the machine tool code language in code files that may be stored in the memory **304**.

[0086] In some embodiments, the communication network **316** may comprise suitable logic, circuitry, and interfaces that may be configured to facilitate communication of data between different components, systems and/or sub-systems in a computing environment that includes the system **300** and other devices, such as machine tools. In some embodiments, the communication network **316** includes the Internet, a local area network (LAN), or any type of network of one or more computers that communicatively couples multiple computers, to name only a few examples. The communication data may be transmitted or received via at least one communication channel of a plurality of communication channels. The communication channels may include, but are not limited to, a wireless channel, a wired channel, or a combination of wireless and wired channels thereof. The wireless or wired channels may be associated with a data standard which may be defined by one of LAN, a Personal Area Network (PAN), a wireless personal LAN (WPLAN), a Wireless Local Area Network (WLAN), a Wireless Sensor Network (WSN), a WAN, and a

Wireless Wide Area Network (WWAN), the Internet, cellular networks, Wireless Fidelity (Wi-Fi) networks, short-range networks (for example, Bluetooth®, WiGig, ZWave®, ZigBee®, IrDA, and the like), and/or any other wired or wireless communication networks or mediums.

[0087] The CNC machine **318** may be configured to process a piece of material, such as a metal, to meet specifications by following coded programmed instructions and without a manual operator directly controlling the machining operation. The coded programmed instructions may be received by the CNC machine **318** in the form of a sequential program of machine control instructions, such as G-code or M-code. Such coded programmed instructions may be executed by the CNC machine **318** to control various machine tools, such as drills, lathes, mills, presses, power saws, and the like, for manufacturing various products, such as various components of the connection assembly mechanism **114**.

[0088] It should be noted that the various components of the connection assembly mechanism **114** may be manufactured automatically by using the machine tools controlled by the CNC machine **318**, as described above, in accordance with some embodiments. However, the disclosure is not so limited, and in accordance with other embodiments, the components of the connection assembly mechanism **114** may be manufactured manually by using hand tools as well, without any deviation from the scope of the disclosure.

[0089] FIG. **4** illustrates the steps of a method **400** of coupling a wall structure and a foundation structure in accordance with some embodiments of the present disclosure. Although specific operations are disclosed in FIG. **4**, such operations are examples and are non-limiting. In different embodiments, to name only a few examples, the method **400** includes other steps, the sequence of the steps is modified, some steps are omitted, or any combination of these variations may be incorporated. The steps of method **400** may be automated or semi-automated. In various embodiments, one or more of the operations of the method **400** can be controlled or managed by software, by firmware, by hardware, or by any combination thereof. FIG. **4** will be explained in conjunction with the descriptions of FIGS. **1-3**.

[0090] In some embodiments, the method **400** includes processes in accordance with the present disclosure which can be controlled or managed by a processor(s) (e.g., **302**) and electrical components under the control of a computer or computing device including computer-readable media containing computer-executable instructions or code. The readable and executable instructions (or code) may reside, for example, in data storage (e.g., **304**) such as volatile memory, non-volatile memory, and/or mass data storage, as only some examples.

[0091] At a step **402**, the method **400** may include determining a type, quantity, and size of the connection assembly mechanism **114** for coupling the foundation structure **102** with the first wall structure **104** based on a first set of parameters associated with the connection assembly mechanism **114** and a second set of parameters associated with the foundation structure **102** and the first wall structure **104**. In some embodiments, the processor **302** in conjunction with the CAU module **310** may be configured to determine the type, quantity, and size of the connection assembly mechanism **114**. The first set of parameters and the second set of parameters are described in detail in the description of FIG. **3**.

[0092] In some embodiments, the foundation structure **102** may correspond to a monolithic cast-in-place foundation structure that is cast at the construction site only. However, the first wall structure **104** may be a precast structure that is cast in a controlled environment of the factory setting.

[0093] At a step **404**, the method **400** may include anchoring the threaded connector **206** in the foundation structure **102**. In some embodiments, the threaded connector **206** may correspond to a cast-in-place anchor bolt. In other embodiments, the threaded connector **206** may correspond to a post-installed anchor bolt set onto the anchoring adhesive **220**.

[0094] At a step **406**, the method **400** may include flush-mounting the hollow box section member **202** with the bottom surface of a vertical precast structure, i.e., the first wall structure **104**. In some embodiments, the hollow box section member **202** may be flush-mounted with the internal bottom

edge of the first wall structure **104** such that the hollow box section member **202** is accessible for the mechanical coupling with the foundation structure **102** from an internal side of the first wall structure **104**.

[0095] At a step **408**, the method **400** may include receiving the threaded connector **206** through the threaded cavity at a base of the hollow box section member **202** at the time of installation.

[0096] At a step **410**, the method **400** may include coupling the hollow box section member **202** and the threaded connector **206** using a connection mechanism, thereby coupling the vertical precast structure, i.e., the first wall structure **104**, with the foundation structure **102**. In some embodiments, the hollow box section member **202** and the threaded connector **206** may be mechanically coupled using the mechanical fastener **210** for coupling the first wall structure **104** with the foundation structure **102** at the construction site.

[0097] At a step **412**, the method **400** may include sealing a gap between the hollow box section member **202** flush-mounted with the internal bottom edge of the first wall structure **104** and the top surface of the foundation structure **102** with the sealant **226**.

[0098] Advantageously, by using the connection assembly mechanism **114** as disclosed herein, construction of the enclosure structure **100** is rapid, with the desired structural integrity and strength. Further, as described above, this process of construction provides stability for the enclosure structure **100**, making it durable and weatherproof, among other such advantages mentioned in detail above.

[0099] The terms “comprising,” “including,” and “having,” as used in the specification herein, shall be considered as indicating an open group that may include other elements not specified. The terms “a,” “an,” and the singular forms of words shall be taken to include the plural form of the same words, such that the terms mean that one or more of something is provided. The term “one” or “single” may be used to indicate that one and only one of something is intended. Similarly, other specific integer values, such as “two,” may be used when a specific number of things is intended. The terms “preferably,” “preferred,” “prefer,” “optionally,” “may,” and similar terms are used to indicate that an item, condition, or step being referred to is an optional (not required) feature of the disclosure. The term “connecting” includes connecting, either directly or indirectly, and “coupling,” including through intermediate elements.

[0100] The disclosure has been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope of the disclosure. It will be apparent to one of ordinary skill in the art that methods, devices, device elements, materials, procedures, and techniques other than those specifically described herein may be applied to the practice of the disclosure as broadly disclosed herein without resort to undue experimentation. All art-known functional equivalents of methods, devices, device elements, materials, procedures, and techniques described herein are intended to be encompassed by this disclosure. Whenever a range is disclosed, all subranges and individual values are intended to be encompassed. This disclosure is not to be limited by the embodiments disclosed, including any shown in the drawings or exemplified in the specification, which are given by way of example and not of limitation. Additionally, it should be understood that the various embodiments of the building blocks described herein contain optional features that may be individually or together applied to any other embodiment shown or contemplated here to be mixed and matched with the features of that building block.

[0101] While the disclosure has been described with respect to a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that other embodiments may be devised which do not depart from the spirit and scope of the disclosure as disclosed herein.

Claims

- 1.** A connection assembly mechanism comprising: a threaded connector configured for embedding in a foundation structure; a hollow box section member having a base including a threaded cavity configured for receiving the threaded connector, the hollow box section member configured for flush mounting with an internal bottom edge of a vertical precast structure; a set of elongated connectors positioned at opposite vertical sides of the hollow box section member, wherein each of the elongated connectors is configured for embedding within the vertical precast structure; and a connection mechanism configured for coupling the hollow box section member and the threaded connector, thereby coupling the vertical precast structure and the foundation structure.
- 2.** The connection assembly mechanism according to claim 1, wherein the connection mechanism corresponds to a mechanical fastener.
- 3.** The connection assembly mechanism according to claim 1, further comprising a plate member configured for distributing a load under the mechanical fastener when the vertical precast structure is affixed with the foundation structure at a construction site, wherein the plate member includes a cavity configured for receiving the threaded connector.
- 4.** The connection assembly mechanism according to claim 1, wherein the hollow box section member is further configured for flush-mounting with the internal bottom edge of the vertical precast structure such that the hollow box section member is accessible for the coupling with the threaded connector from an internal side of the vertical precast structure.
- 5.** The connection assembly mechanism according to claim 1, wherein the threaded connector corresponds to a cast-in-place anchor bolt.
- 6.** The connection assembly mechanism according to claim 1, wherein the threaded connector corresponds to a post-installed anchor bolt set onto anchoring adhesive.
- 7.** The connection assembly mechanism according to claim 1, wherein the foundation structure corresponds to a monolithic cast-in-place foundation structure.
- 8.** The connection assembly mechanism according to claim 1, further comprising a sealant sealing a gap between the hollow box section member flush-mounted with the internal bottom edge of the vertical precast structure and a top surface of the foundation structure.
- 9.** A connection method, comprising: embedding a threaded connector in a foundation structure; flush-mounting a hollow box section member with an internal bottom edge of a vertical precast structure, wherein a set of elongated connectors, positioned at opposite vertical sides of the hollow box section member, is embedded within the vertical precast structure; receiving the threaded connector through a threaded cavity at a base of the hollow box section member; and coupling the hollow box section member and the threaded connector using a connection mechanism, thereby coupling the vertical precast structure with the foundation structure.
- 10.** The connection method according to claim 9, further comprising determining a type, quantity, and size of a connection assembly mechanism for coupling the foundation structure and the vertical precast structure based on a first set of parameters associated with the connection assembly mechanism and a second set of parameters associated with at least one of the foundation structure and the vertical precast structure.
- 11.** The connection method according to claim 10, wherein the first set of parameters includes material specifications of the connection assembly mechanism.
- 12.** The connection method according to claim 11, wherein the material specifications of the connection assembly mechanism correspond to at least tensile strength and hardness of the connection assembly mechanism.
- 13.** The connection method according to claim 10, wherein the second set of parameters includes at least one of a location, an orientation, and a weight of at least one of the foundation structure and the vertical precast structure and environmental conditions.
- 14.** The connection method according to claim 9, wherein the hollow box section member and the threaded connector are mechanically coupled using a mechanical fastener for coupling the vertical

precast structure with the foundation structure at a construction site.

15. The connection method according to claim 14, further comprising distributing a load under the mechanical fastener using a plate member when the vertical precast structure is affixed with the foundation structure at the construction site.

16. The connection method according to claim 9, wherein the hollow box section member is flush-mounted with the internal bottom edge of the vertical precast structure such that the hollow box section member is accessible for the mechanical coupling with the foundation structure from an internal side of the vertical precast structure.

17. The connection method according to claim 9, wherein the threaded connector corresponds to a cast-in-place anchor bolt.

18. The connection method according to claim 9, wherein the threaded connector corresponds to post-installed anchor bolt set onto anchoring adhesive.

19. The connection method according to claim 9, wherein the foundation structure corresponds to a monolithic cast-in-place foundation structure.

20. The connection method according to claim 9, further comprising scaling a gap between the hollow box section member flush-mounted with an internal bottom edge of the vertical precast structure and a top surface of the foundation structure with a scalant.
